

Mental-map: Representing Driver Situational Awareness with Attention-driven Knowledge Graph

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Abstract—Understanding Situational Awareness Levels 2 and 3 requires modelling both drivers’ visual attention and the temporal evolution of scene semantics, as existing driver monitoring systems capture object-level perception but lack mechanisms for representing the spatio-temporal structure. We present mental map, a real-time system that fuses eye-gaze fixations with a dynamic directed knowledge graph to construct a persistent representation of driver awareness. The system integrates YOLOv8 and SAM2 for continuous object detection and tracking, Qwen2.5-VL-7B for semantic hazard triage and relationship reasoning, and a ghost-node memory model that simulates short-term cognitive persistence for temporarily occluded objects. Evaluated across seven naturalistic driving environments from the Look Both Ways dataset, Mental map achieves an aggregate precision of 97.18% and an F1-score of 87.53%, with peak performance observed under overcast, suburban autumn conditions (precision and recall both reaching 95.83%), with performance being comparatively weaker in low-light nighttime settings, where recall falls to 46.43%. We further demonstrate its practical value through an AI coaching module powered by Gemini 2.5 Flash, which generates real-time, context-aware safety interventions grounded in the evolving state of the knowledge graph.

Index Terms—Situational awareness, mental map, eye-gaze tracking, cognitive modelling, knowledge graphs, spatio-temporal reasoning, multimodal fusion, vision-language models, real-time systems

I. INTRODUCTION

Situational awareness (SA) is defined as an individual’s state of “knowing what is going on” within a complex environment [1], [2]. Formally, SA is conceptualised through Mica Endsley’s widely adopted three-level model: the perception of critical elements in the environment Level 1 (L1), the comprehension of their contextual meaning Level 2 (L2), and the projection of their future status Level 3 (L3) [3]–[5]. In dynamic and safety-critical environments, maintaining a high level of SA is essential because it forms the basis for effective

decision-making and safe task performance [6]. However, the ability to successfully build and update this awareness is heavily dependent on and inherently constrained by the limited capacity of their visual attention and working memory [2]. Consequently, when environmental complexities exceed these visual attentional resources, operators frequently experience breakdowns in SA, such as perceptual failures, distraction, or inattention, which are recognised as primary triggers for human error and severe collisions [3], [7]

To mitigate these breakdowns, recent vehicular safety literature argues for a fundamental paradigm shift: moving away from traditional, reactive ‘distraction prevention’ thresholds (such as fixed eyes-off-road durations) toward continuous, context-aware ‘driver attention management and support’ (Reimer et al., 2026). Under this modern Safe Systems approach, an in-vehicle assistant must actively evaluate whether a driver’s dynamic visual engagement is semantically aligned with the contextual demands and risks of the surrounding roadway.

Achieving SA L2, and potentially the deeper understanding associated with L3, is critical in both manual and automated driving, as it enables environmental elements to be interpreted as contextually relevant hazards rather than merely passive objects. Driving is a constantly changing task: drivers must continually balance what they expect to happen, based on experience, with how they react to sudden, attention-grabbing events on the road [8]. The driving context itself, road layout, traffic density, surrounding vehicles, and environmental conditions shape this coordination by setting the perceptual and cognitive demands a driver must meet [9]. Experienced drivers adjust their scanning strategies accordingly, widening their visual search in more complex environments so they can build accurate mental models and anticipate hazards [10], [11].

This allocation of visual attention is often explained through the SEEV system of salience, effort, expectancy, and value [12]. When cognitive load rises, or a secondary task competes

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for visual attention, scanning tends to narrow, weakening SA and impairing how well critical contextual cues are processed [13], [14]. In short, SA Levels 2 and 3 emerge from the integration of contextual understanding, semantic relationships between objects, and their relevance to visual attention, together with the ability to project how the scene may evolve over time.

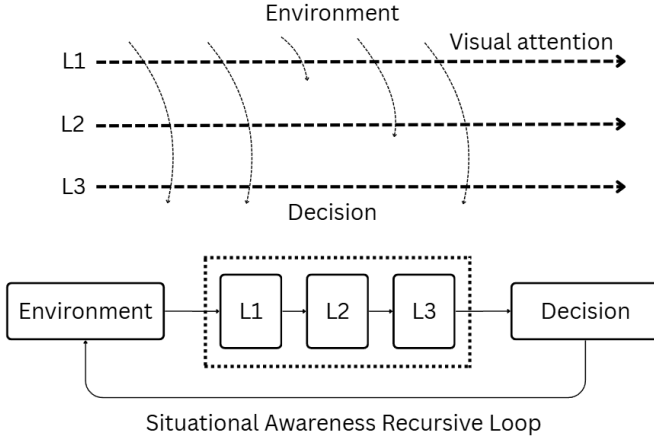


Fig. 1: The figure illustrates the recursive SA loop, where the driver first perceives the environment, processes information through visual attention at L1, and subsequently progresses through higher cognitive layers such as L2 and L3 before making decisions. In our research, we model the environmental representation using a dynamic knowledge graph and capture visual attention through gaze behaviour. By fusing these two modalities into a mental map, we are able to provide insights into the underlying cognitive processes associated with L2 and L3 SA.

Neither component is sufficient on its own: Perception alone, without gaze information, cannot separate what a driver is attending to from what they are ignoring. Gaze alone, without contextual understanding, cannot determine whether something is important or risky. And prediction without grounded perception lacks a reliable basis for anticipating how a situation will evolve. Integrating these components requires reasoning across both spatial and temporal dimensions [15].

The spatial dimension captures the locations of objects relative to the driver’s field of view, demanding that drivers construct a mental model of an unstructured environment, mapping road layout, ego-vehicle position, and the proximity of surrounding traffic participants within their functional field of view [16], [17], which in turn requires the active use of dynamic spatial working memory to monitor shifting locations across multiple visual planes such as the forward windshield and mirrors [18].

The temporal dimension captures how objects enter, persist within, and leave driver awareness over time, as critical cues such as a pedestrian’s trajectory or another vehicle’s maneuvering intent appear at highly variable times before an action occurs [19], with visual attention itself bound by

time, consisting of transient, automatic orienting responses that must transition into perception for an object to be noticed [20]. Drivers do not process scenes uniformly. Instead, they allocate limited attentional resources to a restricted region of the visual field, while recently observed hazards may remain active in working memory after disappearing from direct view. A robust SA model must therefore represent not only objects that are currently visible and attended, but also objects that were recently relevant and may continue to influence decision making, since isolated, frame-by-frame spatial observations cannot distinguish between a fleeting glance and persistent awareness, making the tracking of spatiotemporal trajectories over time indispensable [21].

Without such spatio-temporal grounding, fused through these spatial constraints and temporal continuities to achieve a higher level of semantic interpretation, a model cannot accurately represent the evolving state of driver awareness throughout a continuous driving sequence [22]. This challenge motivates the central research question of this work:

How can a persistent, real-time representation of driver perception be constructed from L1 visual attention patterns, such that temporal and semantic structures in the environment can be used to surface indicative patterns of SA Levels 2 and 3?

To answer this, we propose mental map, an architectural data-fusion layer designed to unite visual attention streams with environmental semantics. Rather than treating scene parsing and eye-gaze tracking as isolated computational blocks, mental map establishes a unified, graph-based representation of driver awareness structured to natively support downstream vehicle telemetry integration, adaptive safety interventions and Human-Machine Interface (HMI)-based feedback mechanisms

The realization of this system is achieved through four core objectives: (1) develop a gaze-to-scene mapping pipeline using a calibrated visual attentional field-of-view model (2) integrate YOLOv8 [23] and SAM2 [24] for continuous object detection and tracking (3) using Qwen2.5-VL-7B [25] for hazard prioritisation and semantic relationship reasoning and (4) construct a persistent knowledge graph and temporal visualisation to model driver awareness.

The contributions can be summarised as follows:

- The implementation of the mental map system, which operationalises driver SA as a continuous temporal process by fusing gaze fixations into a dynamic knowledge graph.
- Fixation detection (dispersion-threshold + parafoveal field-of-view model) and a temporally persistent knowledge graph with node memory to represent attended, known, and missed objects.

Another contribution is the development of an AI-based driver coaching application that integrates raw gaze behaviour, semantic scene interpretation, and SA estimation into an end-to-end pipeline for real-time intervention-based feedback, forming an HMI layer for driver assistance and behavioural guidance. Following the creation of the mental map, the AI coaching pipeline is quantitatively evaluated and demonstrated

through a real-time driver coaching system that generates intervention-based feedback using Gemini 2.5 Flash.

Beyond coaching, the system also enables emerging technological applications and HMI use cases. In real time, the mental map supports a tiered human-machine interface that provides subtle cues for unattended hazards, escalates to warnings for critical missed threats, and suppresses unnecessary alerts when hazards are already attended, enabling context-aware intervention. Over longer periods, aggregated node states enable quantitative metrics such as gaze-to-hazard latency, attention coverage, and dwell adequacy for behavioural analysis. The evolving graph also supports workload estimation by identifying high-demand scenes and filtering irrelevant stimuli to reduce cognitive load, while temporal node tracking enables predictive risk estimation (e.g., closing speed and time-to-event), shifting the system from reactive perception to anticipatory hazard detection. The code for the mental map and AI coaching demonstration is available at: <https://github.com/AI-for-Road-Safety/mental-map>

II. METHOD

A. Dataset

This study utilises a high-quality naturalistic driving dataset to support robust fixation and SA modelling. Four candidate datasets, including LOTVS DA [26], DR eye VE [27], MEDIRL EyeCar [28], and Look Both Ways LBW [29], were systematically evaluated based on data quality, operational domain coverage, and availability of gaze and tracking annotations. The LBW dataset was selected due to its high-fidelity 3D gaze measurements validated using eye-tracking glasses, large-scale data volume of approximately 300 GB, and a recent publication. It contains recordings from 28 participants across urban, rural, and residential driving environments under diverse weather and lighting conditions, yielding approximately 7 hours of driving data.

The processing pipeline uses only the scene IMS forward-facing video frames and gaze info, 2D and 3D gaze and eye location data streams. Temporal alignment is performed using timestamp matching to ensure synchronisation between gaze and visual data. The dataset is further downsampled from 15 Hz to 5 Hz to reduce computational load while preserving temporal consistency. In addition, all scene frames are reorganised into a sequential indexing format to standardise processing order and ensure reproducibility.

To address limitations in conventional fixation modelling approaches, such as strict point-to-bounding-box matching or uniform spatial assumptions, the proposed mental map introduces a field-of-view approximation for driver attention modelling. This is implemented using a 60-pixel parafoveal tolerance derived from the LBW camera configuration of 1920-pixel width and 75 to 90 degree horizontal field of view, corresponding to approximately 24 pixels per degree of visual angle. A parafoveal range of 2.5° is used, corresponding to a 60-pixel perceptual radius, following [30].

For each frame, fixation assignment is determined by computing the minimum Euclidean distance between the gaze

coordinate (g_x, g_y) and the nearest boundary of each object bounding box (b_x, b_y) : $D = \sqrt{(g_x - b_x)^2 + (g_y - b_y)^2} \leq 60$. If this condition is satisfied, the corresponding object is registered as a valid fixation via its unique tracking ID.

B. Architecture

The proposed mental map operationalises SA assessment in naturalistic driving through a four-step pipeline built around a directed knowledge graph, which represents the EGO vehicle as a central node linked to surrounding objects via directed edges that capture asymmetric spatial and causal relationships. The process begins with data loading and gaze fixation detection using the IDT algorithm, alongside initialisation of a model suite comprising YOLOv8 for detection, SAM2 for segmentation, Qwen2.5-VL for semantic reasoning; Qwen 2.5 VL (7B) was selected for contextual scene labeling after evaluation against alternatives including LLaVa, Video-LLaMa [31] and Intern VL3 [32], owing to its superior overall performance [33].

During the Semantic Pass Loop, frames undergo semantic threat triage, categorising objects as Imminent, Latent, or Point of Interest, while an object-proximity filter constrains relational computation to hazard-relevant clusters to reduce overhead. These results feed into a Temporal Object Tracker. The subsequent Temporal Pass Loop maintains consistency by generating object masks via SAM2 and updating the Directed Knowledge Graph, supported by ghost nodes that retain unobserved objects within a rolling five-frame window to model the short-term memory a human driver would apply to temporarily obscured hazards.

Safety-critical integrity throughout this process is preserved via a closed-loop verification mechanism, in which three independent candidate-generation loops and a secondary judge prompt detect and correct structural contradictions or tracking errors in the graph.

After the mental map is created, we evaluate one of its potential applications through AI coaching. Based on the mental map, the system performs SA scoring by classifying driver engagement as Low, Medium, or High. It also assesses the quality of Gemini-generated safety advice by comparing it against the knowledge graph using a five-level rating scale ranging from Very Good to Very Bad. In addition, an alert-suppression module cross-references driver gaze fixations with graph-identified hazards to issue warnings only when they are genuinely necessary. The outputs from these modules are integrated into a comprehensive visual interface that demonstrates the potential of the mental map for real-time AI coaching and driver feedback. Details of each module are provided below:

1) *Semantic threat triage*: the scene frame with its mapped tracking labels is fed into the Vision Language Model (VLM) together with a prompt that categorises each object into one of three distinct threat levels: Imminent which refers to objects on immediate collision or direct risk trajectories with the ego vehicle, Latent which represents potential or context-dependent risks that may become relevant depending on future motion or environmental changes, and POI or Points of Interest

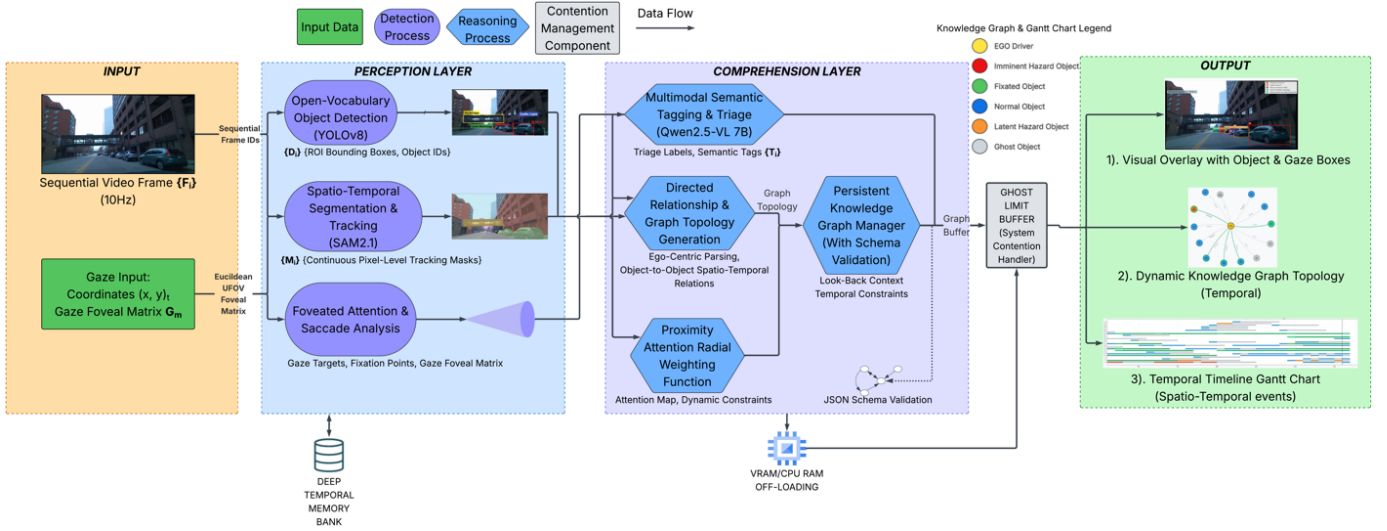


Fig. 2: Overview of the mental map architecture organised into four stages linked by a continuous memory loop. The Input stage ingests sequential video frames (10 Hz), gaze coordinates, and a foveal gaze matrix from the Deep Temporal Memory Bank. The Perception Layer performs object detection (YOLOv8), spatio-temporal tracking and segmentation (SAM2.1), and foveated attention analysis to produce bounding boxes, masks, and fixation points. The Comprehension Layer applies multimodal semantic tagging (Qwen2.5-VL-7B), constructs ego-centric relational graphs, and computes a proximity-weighted attention function, with outputs managed by the Persistent Knowledge Graph Manager to ensure temporal consistency. A Ghost Limit Buffer handles compute and memory offloading between layers. The Output stage renders visual overlays, the dynamic knowledge graph (Temporal), and a temporal Gantt chart of spatio-temporal events.

which are neutral objects that contribute to general scene understanding but do not pose any driving threat.

2) *Contextual scene labelling*: early versions of the system relied on the LLaVA model for contextual scene labelling. However, testing and technical reviews revealed that LLaVA suffered from performance degradation and inconsistent outputs when processing large, complex driving scenes. To resolve these issues, we evaluated several alternatives of similar size, including Video-LLaMa and InternVL3-9B. While InternVL3-9B and Video-LLaMa offer strong consistency they both lost out to Qwen 2.5VL 7B in alignment, physics and commonsense [33]. Due to these factors, Qwen 2.5 VL (7B) was selected as the optimal architecture for this system. Its capability in precise object grounding and video understanding directly addresses the requirements for high-accuracy semantic labelling [25]. This transition has significantly improved both scene-labelling reliability and processing speeds, resulting in a measurable increase in the quality of data ingested by the downstream knowledge graph.

3) *Object-proximity filter*: reduces computational overhead by avoiding relationship estimation across all visible nodes. The VLM is constrained to focus only on tracks identified in the previous stage as hazards. It then analyses the spatial context around these selected IDs and computes relational structures for object clusters located within their immediate neighbourhood.

4) *Directed knowledge graph*: encodes the EGO vehicle as a central node and links surrounding objects (e.g., vehicles, pedestrians, hazards) through directed edges that reflect asym-

metric relationships. This directionality is important because it preserves causal or spatial influence, such as a nearby vehicle affecting the EGO’s risk state without the reverse necessarily being true, which supports reasoning over dynamic interactions and evolving threat levels within the driving environment.

5) *Ghost nodes*: are unobserved objects maintained in the active knowledge graph for a rolling window of 5 frames. Ghost nodes were added to model human short-term memory, as a driver would not instantly forget about a recently observed hazard simply because it was temporarily obscured.

6) *Safety-critical*: the system incorporates object detection and tracking loops that continuously feed a local VLM for threat prioritisation within a closed-loop verification mental map. Semantic relationships produced by the VLM are constrained through a directed knowledge graph, where guardrails are applied to detect and remove spatial inconsistencies and tracking errors in real time.

C. Implementation details

The mental map was implemented in Python using PyTorch and optimised for GPU memory efficiency, automatic mixed precision using bfloat16, garbage collection, and CUDA cache clearing to prevent out-of-memory errors. Gaze fixations were extracted using a dispersion-threshold algorithm requiring a maximum dispersion of 50 pixels and a minimum duration of 100 ms at a 30 Hz sampling rate. Nearby fixations within a 30-pixel Euclidean distance were merged before being projected into 3D space using synchronised depth maps and camera intrinsics ($c_x = 384$, $c_y = 256$, $f_x = 500$, $f_y = 500$).

Prompt 1	Prompt 2
Look at these specific tracked objects currently in the scene: {tids_str}.	Look at these specific tracked objects: {tids_str}.
Describe what each object is doing and its location relative to the camera car (EGO). (e.g., driving ahead, oncoming traffic on the left, parked on the right).	Identify if any of these objects are near each other (not the EGO car). If they are near another object, describe their relation.
Keep it under 5 words per object.	If none are near others, say NONE.
Output format: REL: EGO->T0: parked on right ...	Output format: REL: T2->T3: following behind ...

TABLE I: Prompts 1 and 2 extract object behaviour and spatial relationships from tracked scene elements. These prompts are utilised by the Qwen2.5-VL-7B-Instruct vision-language model to perform semantic scene reasoning every 10 frames. Prompt 1 facilitates ego-centric parsing by instructing the model to succinctly describe the action and location of tracked objects relative to the camera (EGO) vehicle. Prompt 2 conducts inter-object relationship analysis by evaluating the spatial proximity and relational dynamics among external objects, explicitly excluding the EGO vehicle. Both prompts mandate a strict relational output format (e.g., REL: EGO → T0: parked on right), which is subsequently constrained using regular-expression parsing to ensure structured data representations for the system’s directed knowledge graph.

Object detection was performed using YOLOv8s with a confidence threshold of 0.5, while temporal tracking used a custom tracker with a minimum intersection over union (IoU) threshold of 0.3. Pixel-level object propagation was achieved using the SAM2.1 Video Predictor, initialised with YOLO-derived object prompts (IDs 0–99) and gaze prompts (IDs 100+). Gaze prompt radii were dynamically scaled according to fixation duration using $20 + \text{Duration in ms}/100$

Semantic scene reasoning was conducted using Qwen2.5-VL-7B-Instruct loaded in 4-bit quantised form with bfloat16 computation. Every 10 frames, the model performed three sequential tasks consisting of hazard triage, object-to-EGO relationship estimation, and inter-object relationship analysis. Outputs were constrained through regular-expression parsing to ensure structured representations.

The mental map was implemented with a 20-frame memory window. Untracked objects transitioned into ghost nodes that persisted for 5 frames before removal. Node states were updated according to object intersections with a 60-pixel

foveal region surrounding the driver’s gaze, producing cognitive state labels of LOOKING, REMEMBERED, MISSED, OBSERVED, and GHOST.

Driver coaching and SA assessment were generated using Gemini-2.5-Flash, constrained to JSON outputs. The advisory module generated three candidate warnings and selected the most actionable response through a secondary judging stage. SA was subsequently classified as Low, Medium, or High using the generated advice, knowledge graph state, and raw gaze data.

III. RESULTS

The system achieves an aggregate precision of 97.18%, an aggregate recall of 79.62%, an F1-score of 87.53%, and a mean Average Precision (mAP) of 65.72%. However, prioritising precision also introduced a critical recall vulnerability in low-contrast environments. This trade-off was directly shown during low-light nighttime testing (S262, Table III), where recall dropped significantly to 0.464, dragging the overall system recall down to 79.62%. This imbalance was further reflected in the mean Average Precision (mAP), which fell to 65.72% likely due to the strict confidence filtering restricting the performance across varied detection thresholds compared to conventional baselines like YOLOv8 (92.8% mAP).

TABLE II: Object detection benchmarking comparing the proposed mental map architecture against five baselines (SSD, Faster R-CNN, RT-DETR-R18, YOLOv8, YOLOv10) across 70 driving frames in seven operational environments.

Model	Precision	Recall	F1-Score	mAP
SSD	89.7%	64.4%	69.0%	72.4%
Faster R-CNN	86.5%	74.3%	79.9%	88.2%
RT-DETR-R18	84.7%	80.5%	82.5%	90.4%
YOLOv8	92.9%	88.1%	85.3%	92.8%
YOLOv10	93.0%	85.1%	89.0%	92.2%
Proposed architecture	97.18%	79.62%	87.53%	65.72%

TABLE III: Quantitative object detection performance across seven environmental operational domains using Precision, Recall, and F1-score metrics. These measures evaluate localisation accuracy and reveal the system’s susceptibility to increased false negatives under nighttime conditions.

Scene Data	Precision	Recall	F1-Score
S062 (Overcast/Urban)	0.9524	0.7843	0.8602
S111 (Clear/Urban)	1.0000	0.7273	0.8421
S173 (Sunny/Urban)	1.0000	0.8125	0.8966
S191 (Clear/Snowy/Suburban)	0.9744	0.8085	0.8837
S242 (Clear/City/Urban)	0.9762	0.8542	0.9111
S262 (Night/Urban)	1.0000	0.4643	0.6341
S282 (Overcast/Autumn/Suburban)	0.9583	0.9583	0.9583
Overall average	0.9718	0.7962	0.8753

System performance was evaluated on a five-tier scale based on errors per frame, from “Very High” accuracy (0 errors) to “Very Low” accuracy (7+ errors). Errors were recorded when the system lost track of an object, failed to update its

temporal history, or produced incorrect VLM-based outputs. This manual assessment was necessary because automated ground-truth metrics cannot directly evaluate the validity of such cognitive graph structures.

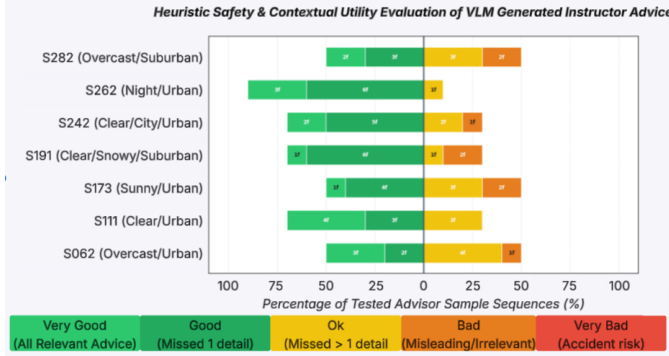


Fig. 3: A 100% stacked diverging bar chart illustrates error-frequency distributions across video frames to assess look-back window stability and graph-state clutter across seven distinct scene profiles. It evaluates the overall quality and contextual utility of safety advice generated by the system’s Gemini AI coaching module by cross-referencing the generated text against the underlying knowledge graph. The performance of the AI advisor is quantified using a five-level rating scale: “Very Good” (All Relevant Advice), “Good” (Missed 1 detail), “Ok” (Missed > 1 detail), “Bad” (Misleading/Irrelevant), and “Very Bad” (Accident risk). Results are distributed across various environmental conditions, demonstrating how the heuristic safety of the generated interventions holds up in scenarios ranging from clear urban streets to low-light nighttime settings.

Six representative scenes were chosen for detailed analysis, each highlighting a specific failure mode across the perception, memory, and language-generation components of the pipeline. These range from missed detections of small, distant hazards that fail to clear the confidence threshold, through graph-level artefacts produced by the interaction between ghost-node persistence and an unconstrained proximity prompt, to boundary effects introduced by the circular parafoveal threshold itself, and finally to cases where the system’s natural-language output becomes temporally stale or structurally decoupled from the tracked scene entities it is meant to describe. The analysis of the six scenes is presented as follows:

A consistent localisation error was identified across multiple testing environments: the systematic missed detection of mid-to-long-distance traffic lights (see Scene 1, Figure 4). In these scenarios, low-resolution, distant pixel clusters failed to cross the pipeline’s strict confidence threshold, causing the system to overlook smaller background hazards.

System audits identified instances where a directed edge looped recursively back onto its own node (e.g., node T0 in Scene 2, Figure 4). This occurred when a target passed beneath a low-contrast concrete overpass, causing the tracking models to temporarily drop their bounding box. Because the mental map’s 5-frame ghost-memory window kept the tracking label

frozen to ensure temporal persistence, the secondary proximity prompt lacked an identity-exclusion constraint. As a result, the VLM evaluated the single saved node against itself and returned a self-referential relationship (T0 to T0).

During sharp manoeuvres, this persistence leak preserved out-of-frame objects, cluttering the NetworkX graph with “zombie” nodes (Scene 3, Figure 4). Furthermore, the lack of an identity-exclusion constraint allowed the VLM to evaluate a single cached node label against itself, generating self-referential tracking loops (T0 to T0).

By implementing a circular vision threshold, the mental map accurately captures human parafoveal tracking behaviour. As verified in Scene 4 (Figure 4), when a driver looks near a target, the system correctly registers their peripheral awareness rather than logging a critical attention lapse. The empirical behaviour of this tracking boundary is highlighted in scene S279 (Scene 4, Figure 4), which captures a dense cluster of road targets surrounding a central gaze coordinate. Under traditional, rigid bounding-box collision logic, the system would have recorded a tracking failure, logging the driver as completely unaware of adjacent road targets. However, the 60-pixel radial threshold successfully tracked these close-proximity objects, correctly registering them as a driver-identified hazard when gaze vectors fell near, but not directly inside, the object boundaries.

In Scene 5 (Figure 4), the model issued an outdated safety warning (“Watch for the pedestrian crossing the street ahead”) after the target pedestrian had already completed the crossing and stepped onto the opposite curb. More critically, Scene 6 (Figure 4) exhibited a direct tracking hallucination, generating the tactical instruction, “Note the parked car on the right... Adjust your path,” when the roadside contained only a standard concrete curb. This error was triggered by extreme light-exposure transitions beneath an overpass, where deep geometric shadows caused false features that the VLM misidentified as a stationary vehicle.

The false warning in Scene 6 (Figure 4) revealed a software loophole that bypassed system filters and appeared on the driver interface. The issue occurred because the generated advice relied on unanchored descriptive text rather than explicit tracking keys (e.g., T0 or T1), leaving the regex-based parser without stable numeric identifiers to match YOLOv8/SAM2 outputs. The unverified string bypassed the alert-suppression logic and triggered the visual alert flag, highlighting a grounding gap between generated outputs and tracked scene entities.

IV. DISCUSSION

The development of the mental map system directly responds to the paradigm shift advocated by Reimer et al. [34], who argue for a transition away from traditional, isolated distraction-prevention metrics toward continuous, context-aware driver attention support. While Reimer et al. outline the conceptual necessity of aligning a driver’s attention with real-time environmental demands, the dynamic knowledge graph system presented here operationalises this concept. By mapping visual fixations directly onto semantically tagged, egocen-

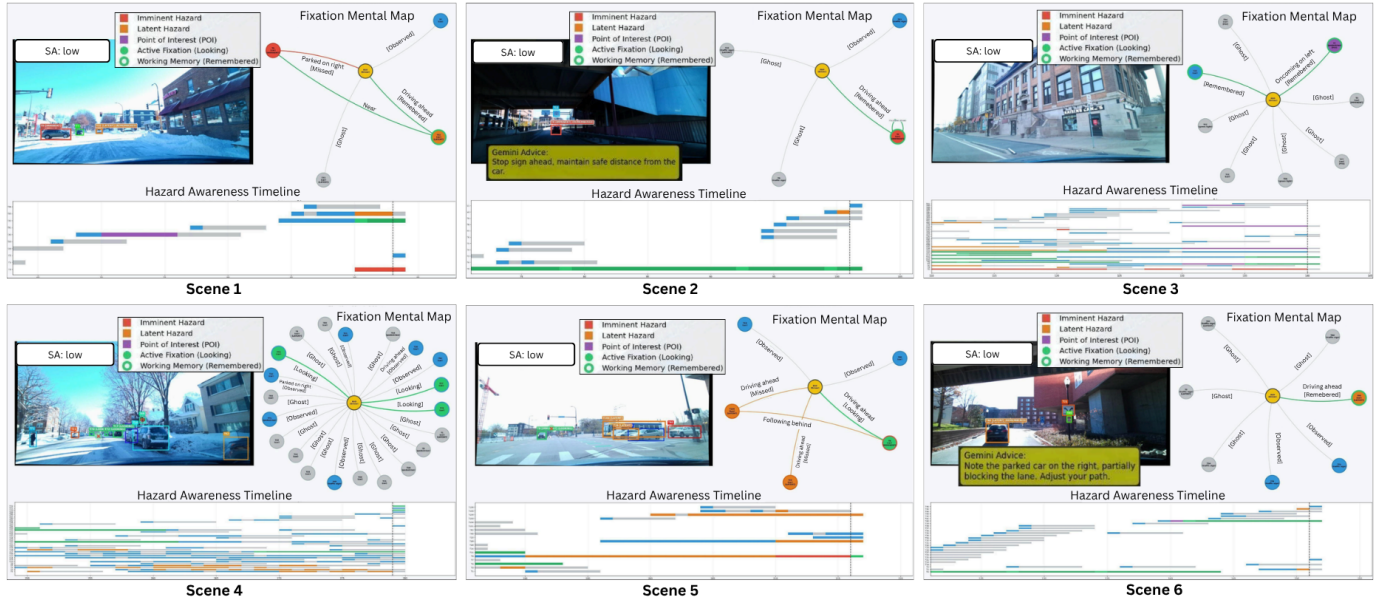


Fig. 4: This figure presents the image view, the knowledge graph (mental map), and the Gantt chart of events. The image shows detected objects with bounding boxes annotated with mental map IDs, fixation status, and a legend for colour coding. It also includes a Gemini module that highlights a missed object by YOLOv8 and indicates a low driver SA score. The knowledge graph on the right represents the driver’s mental map, showing classified objects and their relationships, including contextual edge labels. At the bottom, the Gantt chart maps frame numbers against object IDs to show temporal evolution. Scene 3 illustrates “zombie” nodes, where two current nodes persist in the mental map even though the corresponding objects are no longer visible.

tric relational graphs, our system moves beyond checking if a driver is looking ‘off-road’. It dynamically assesses whether its cognitive model is actively integrated with the surrounding risk landscape. Furthermore, our ghost-node memory mechanism exemplifies the Safe Systems Approach emphasised by Reimer et al. by engineering system resilience around human visual and working-memory constraints, tracking temporarily obscured threats to ensure structural support before an attentional deficit results in a failure to respond.

The mental map’s detection performance reflects a deliberate precision-focused tuning strategy, evaluated on the Look Both Ways dataset. It achieves high precision and F1-score, indicating highly reliable detections, though this reliability is balanced by a measurable reduction in recall. By constraining the model to a narrow target operational window via strict confidence filtering, the system was tuned to minimise false positives, ensuring that flagged hazards corresponded with high fidelity to genuine objects in the scene.

This configuration, however, introduced a corresponding vulnerability to false negatives, with missed detections occurring more frequently than would be expected under a recall-optimised configuration. The effect was most pronounced in low-contrast operating conditions, particularly low-light and nighttime scenes, where reduced visual contrast degraded the confidence scores assigned to genuine hazards, pushing a subset of true positives below the detection threshold. These results are consistent with the system’s underlying design ra-

tionale: in a driver-attention-management context, false alarms were treated as more damaging to operator trust and sustained engagement than occasional omissions of less salient hazards, justifying the conservative bias toward precision observed in the detection outputs.

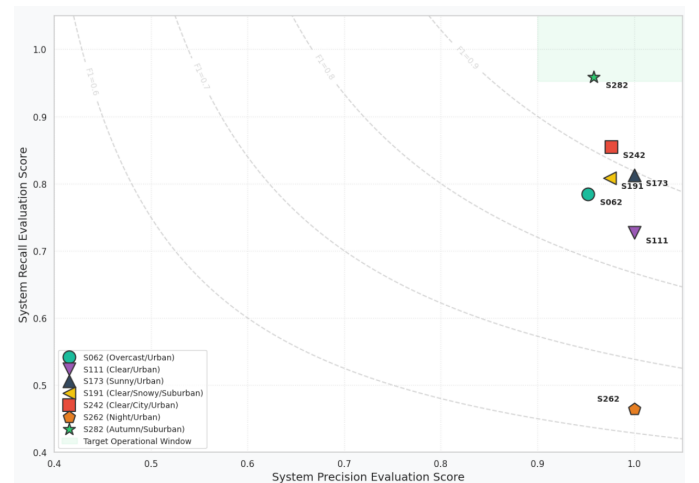


Fig. 5: Detection tuning trade-offs across seven naturalistic driving scenarios. This 2D scatter plot compares precision and recall across seven distinct driving environments, highlighting a clearly defined “Target Operational Window” that illustrates the inherent trade-off in tuning an open-vocabulary YOLOv8 model for high-precision performance.

Several limitations were identified during evaluation, including the difficulty of sustaining consistently high-quality generative outputs across the full range of driving scenarios encountered in the dataset and the continued need for refinement in prompt engineering, particularly for the Gemini-based AI coaching module built upon the mental map. The variability in scene complexity, occlusion patterns, and traffic density meant that coaching outputs occasionally lacked the contextual specificity needed to be immediately actionable, with feedback sometimes proving too generic or, conversely, overly verbose for real-time consumption. This sensitivity to prompt phrasing also introduced inconsistency in tone and structure across sessions, suggesting that further work is needed to stabilise output quality independent of scene-specific variation.

A key limitation of the parafoveal approach is its fixed 60-pixel radius, which is based on a single camera setup and does not scale with depth or vehicle speed. This leads to distorted visual-angle approximations, overestimating peripheral awareness for distant objects and underestimating it for nearby ones.

Stress-testing revealed key failure modes. Recall dropped to 0.464 in low-light conditions, reducing the reliability of hazard suppression in critical scenarios. Rapid intersection manoeuvres caused “zombie node” anomalies, where ghost-memory persistence led to self-referential tracking loops. An unanchored text vulnerability also allowed generated alerts to bypass safety constraints, indicating that reliability depends on strict grounding to verified tracking IDs.

A theoretical limitation is that gaze-based fixation serves only as a proxy for attentional state, approximating rather than fully capturing visual cognition. This choice prioritises interpretability and computational tractability over representational fidelity. Extending the system to SA L3 projection will likely require moving beyond fixation-based proxies toward world-model approaches that enable temporal reasoning about physical and behavioural dynamics, allowing anticipation of how the scene and road users may evolve.

V. CONCLUSION

This research presents the mental map, an under-development tool for representing driver SA with an attention-driven knowledge graph, formed by fusing driver fixation data with a dynamic knowledge graph. The system develops a better understanding of Endsley’s SA Levels 2 and 3 by leveraging L1 visual perception and environmental context as the foundation for interpretation.

Beyond detection and tracking, it demonstrates practical value by maintaining attended, currently known, and missed objects as persistent nodes. This enables tiered hazard alerting that suppresses unnecessary warnings when a driver is already attending to a threat, while escalating cues for genuinely missed hazards.

The structural architecture of the mental map system is inherently modular, serving as the foundational data layer required for end-to-end, proactive attention support systems. Because the system outputs a unified, real-time directed knowledge graph rather than disjointed raw data streams,

it naturally accommodates the integration of broader safety layers. For example, the semantic edges connecting environmental nodes can be directly parameterised in subsequent iterations to include quantitative risk metrics.

The design directly operationalises SA, distinguishing objects a driver has merely perceived from those they understand as contextually relevant risks, and forms the basis for the AI coaching application, where Gemini 2.5 Flash grounds its advisory feedback in the evolving graph state rather than raw detections alone.

Furthermore, by establishing a stable, graph-based representation of driver awareness, this architecture provides the necessary baseline to evaluate gaze uncertainty and dictate context-specific HMI policies, such as alert suppression or proactive steering guidance, ensuring that downstream safety interventions are grounded in an accurate estimation of the driver’s cognitive state.

A key contribution was the effective real-time integration of modern VLMs, Qwen2.5-VL for semantic analysis and knowledge graph construction, and Gemini 2.5 Flash for driving advice, addressing a gap in prior work that was limited to lower-performing models. The system combined YOLOv8 object detection, SAM2 segmentation, and a gaze-based fixation algorithm accounting for dispersion thresholds and minimum duration, demonstrating consistent performance across diverse scenarios, including urban, residential, day and night, and varying weather conditions, achieving a validation precision of 97.18% and an F1-score of 87.53% across seven naturalistic driving environments.

The hazard-relevant structure was reliably captured through the knowledge graph, as shown in recall and mAP (79.62% and 65.72% respectively) and precision, reflecting a deliberate conservative bias suited to alert-suppression use cases where false positives are more costly to driver trust than missed low-salience detections.

Future work will improve the robustness and reliability of the proposed mental map system. Additional sensing modalities, such as infrared cameras and LiDAR, will be integrated to enhance low-light performance and improve depth estimation, while vehicle telemetry data will support more robust object tracking in complex driving manoeuvres. Larger-scale evaluation on datasets such as HAD and HDBD will also be conducted to improve generalisability and alignment with emerging driver monitoring standards. From a reasoning perspective, future work will address hallucinations and information overload caused by dense secondary graph connections by refining which object classes, attributes, and relationships are retained in the knowledge graph. Model upgrades present a further clear opportunity: adopting Gemini 3.5 Flash would enable lower-latency reasoning, closing the gap between perception updates and timely feedback delivery, while integrating Qwen3-VL could enhance spatial and multimodal understanding. Taken together, these advances reinforce the mental map’s role as a persistent, interpretable, and scalable representation of driver SA, paving the way for new technological applications, HMI integration, coaching feedback, and real-time intervention.

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